



PIGMENTED LESIONS OF ORAL MUCOSA -3

Dr. Sajda Khan Gajdhar
MDS, Oral and Maxillofacial Pathology



LECTURE LEARNING OUTCOMES

At the end of the lecture, students should be able to:

Identify and differentiate malignant pigmented lesions in the oral cavity

Classify Malignant Melanoma

Describe clinical, histopathological, and radiographic features of malignant melanoma

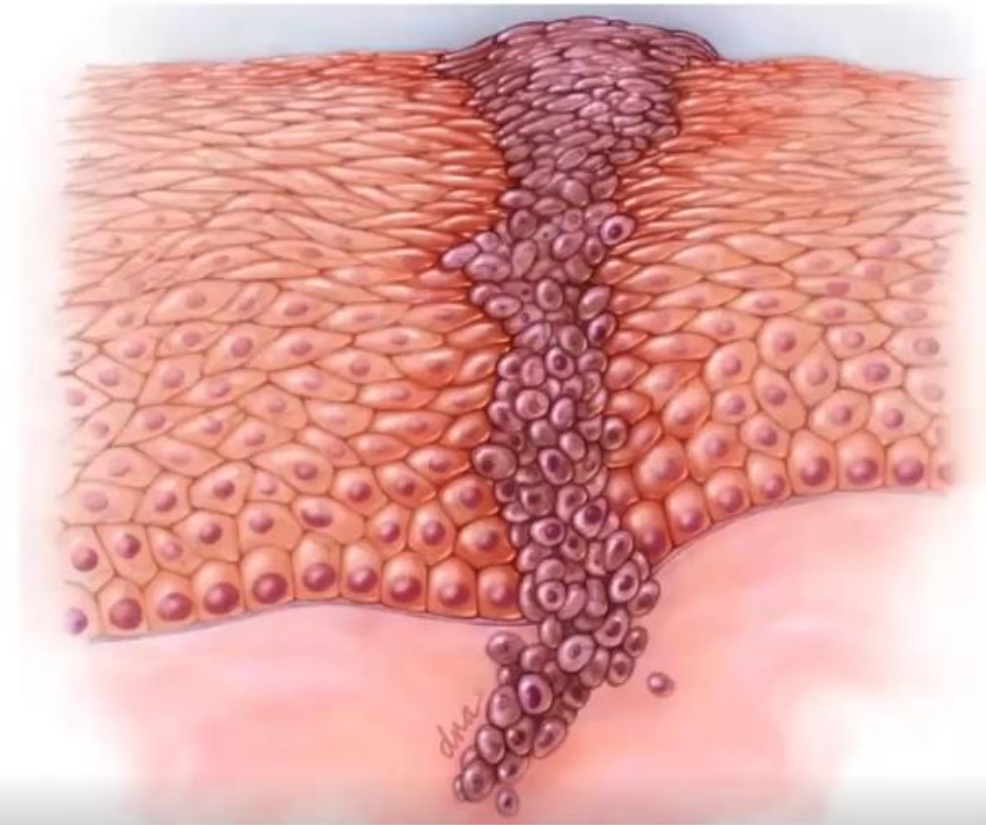
Essential reading:

Regezi & Sciubba: Oral Pathology - Clinicopathologic Correlation; r (6th edn), Chapter 5, pg: 134-148

Shafers textbook oral pathology((7thedn)

MALIGNANT MELANOMA

- Third Most common cancer of Skin common in fair skinned people.
- Occur from pre-existing premalignant lesions or de novo.
- **Etiology:** Sun exposure, Artificial UV Radiations.
- **Predisposing factors:** Socioeconomic Status, Freckles, Melanocytic Nevus, Family history, Xeroderma Pigmentosum.



PHASES OF GROWTH

1. RADIAL (superficial spreading) GROWTH PHASE

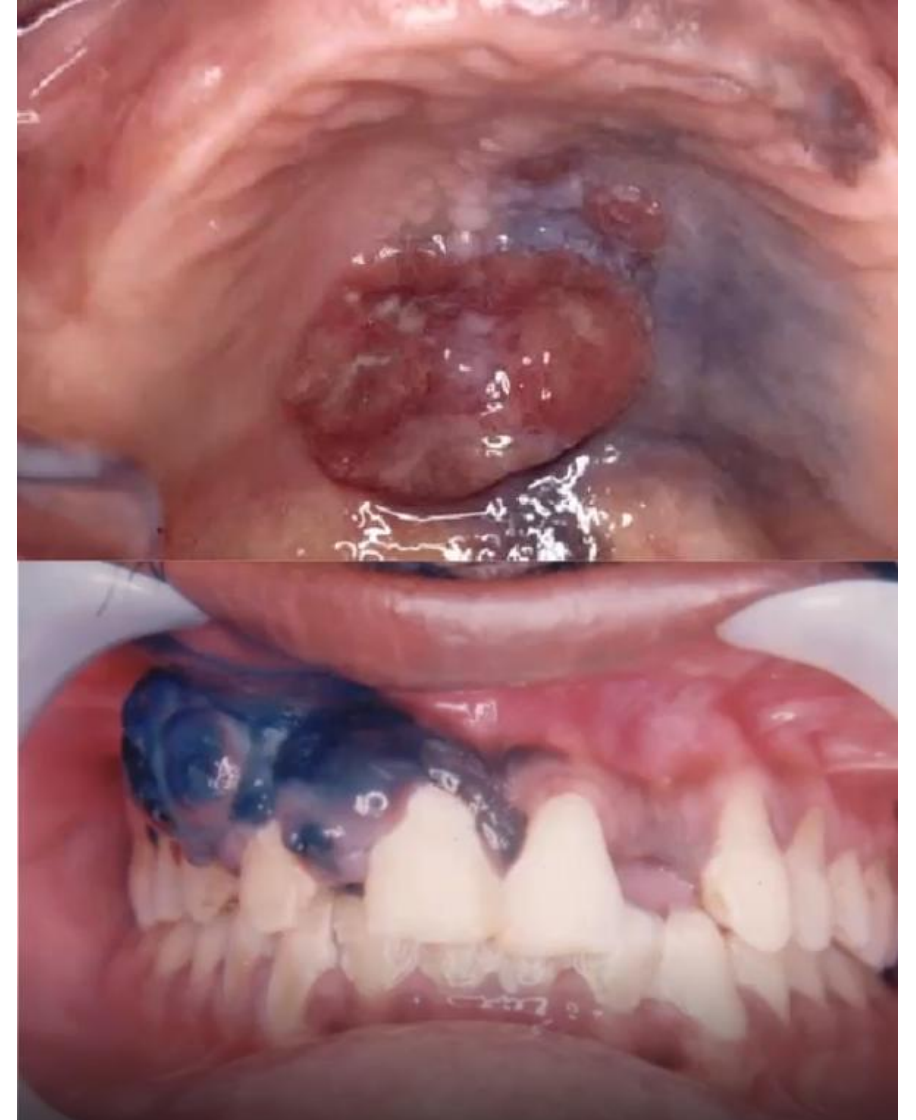
large melanoma cells in the superficial epithelium.

Tumor expands radially

2. VERTICAL GROWTH PHASE

Aggressive growth pattern with deep infiltration, development of palpable nodule.

Nodular type of melanoma has only VGP



Classification of cutaneous melanoma

1. Superficial spreading melanoma
2. Lentigo maligna melanoma
3. Nodular melanoma
4. Acral-lentiginous melanoma
5. Mucosal lentiginous

SALE NAME

Differential Diagnosis: American 'ABCDE' system



SYMMETRY



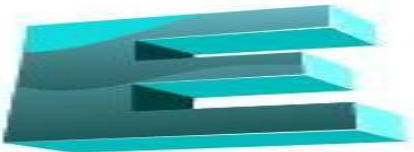
ORDER IRREGULARITY



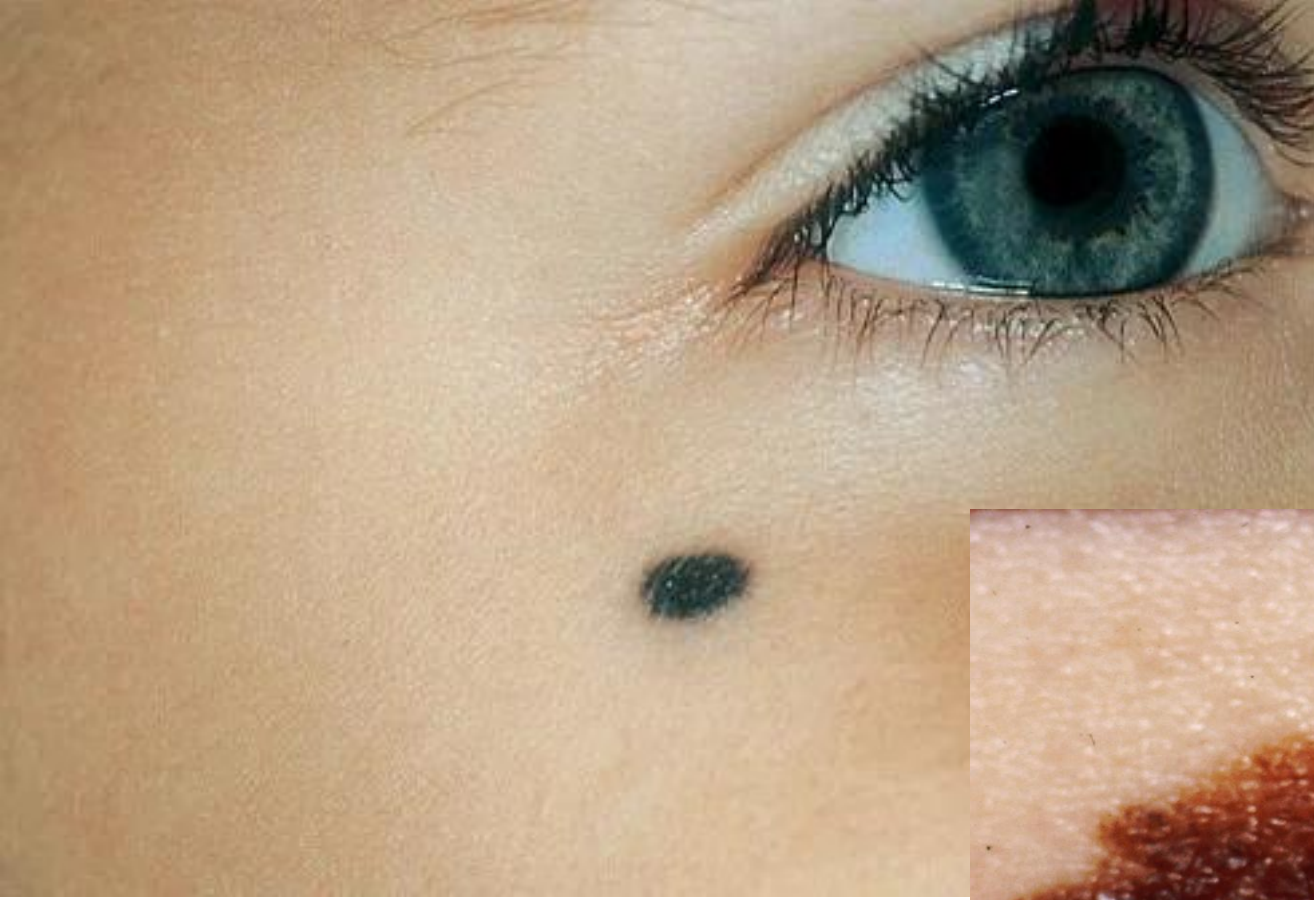
COLOR VARIATION



DIAMETER GREATER THAN 6mm



ELEVATION/Evol



Superficial spreading melanoma



Nodular

- Most aggressive, nodular elevation
- Vertical growth phase

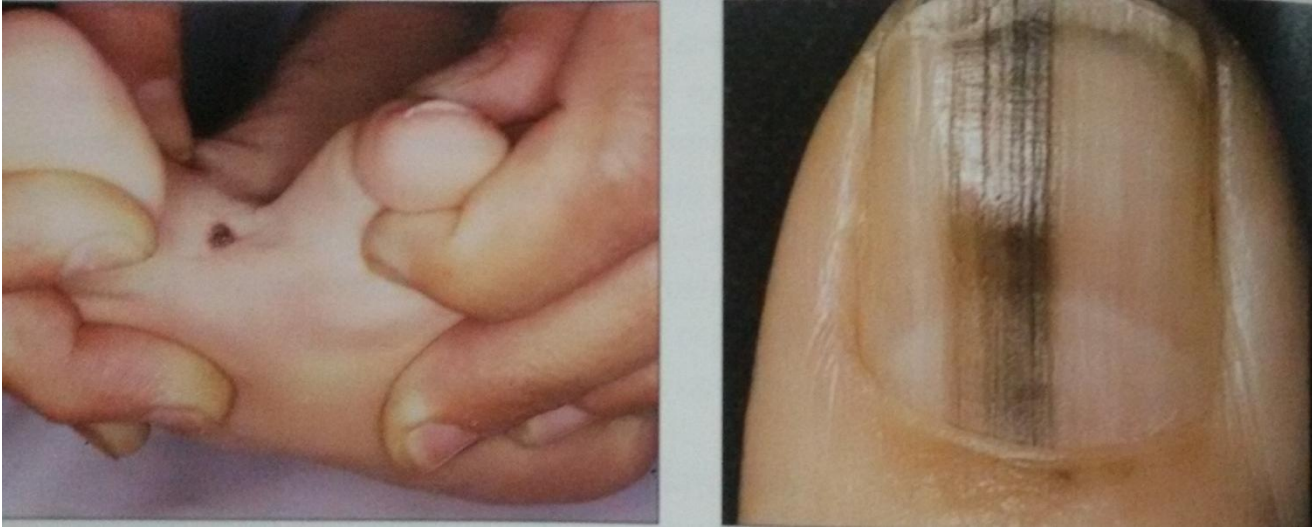


Lentigo maligna Melanoma

- Sun-exposed skin
- Best prognosis



Acral lentiginous Melanoma



- **Very aggressive**
- Commonly arises on palms, soles
- **Pigmentation of nail bed**
- poorer prognosis



Acral Lentiginous

ORAL MALIGNANT MELANOMA

- Rare, 0.5% of all oral malignancies
- Aggressive tumour of melanocyte
- Male > female
- Older age- 4-7 decade of life
- blacks and Asians > whites

ABCDE
RULE

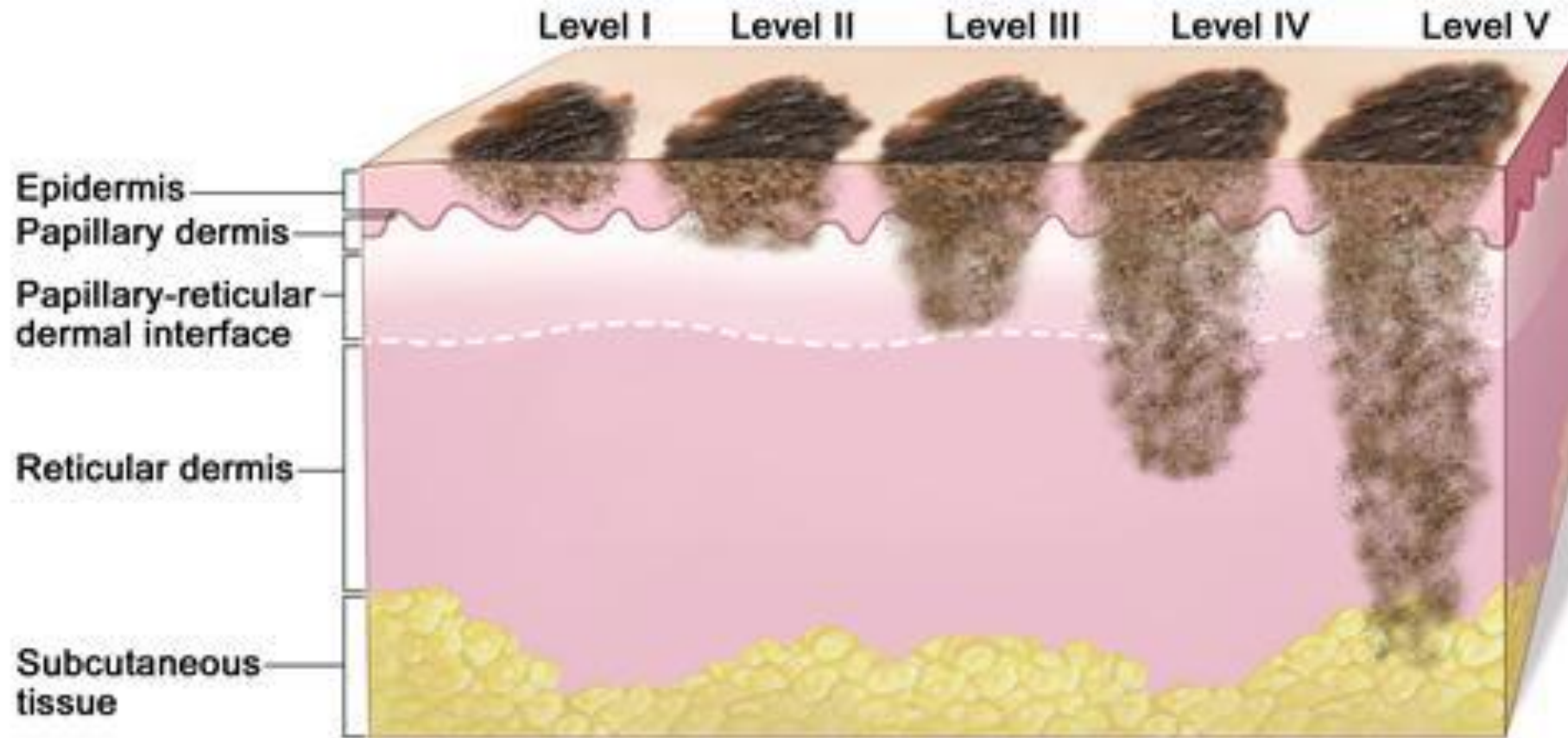


- Early lesions –flat but later become nodular.
- **Site: Hard palate and maxillary alveolus.**



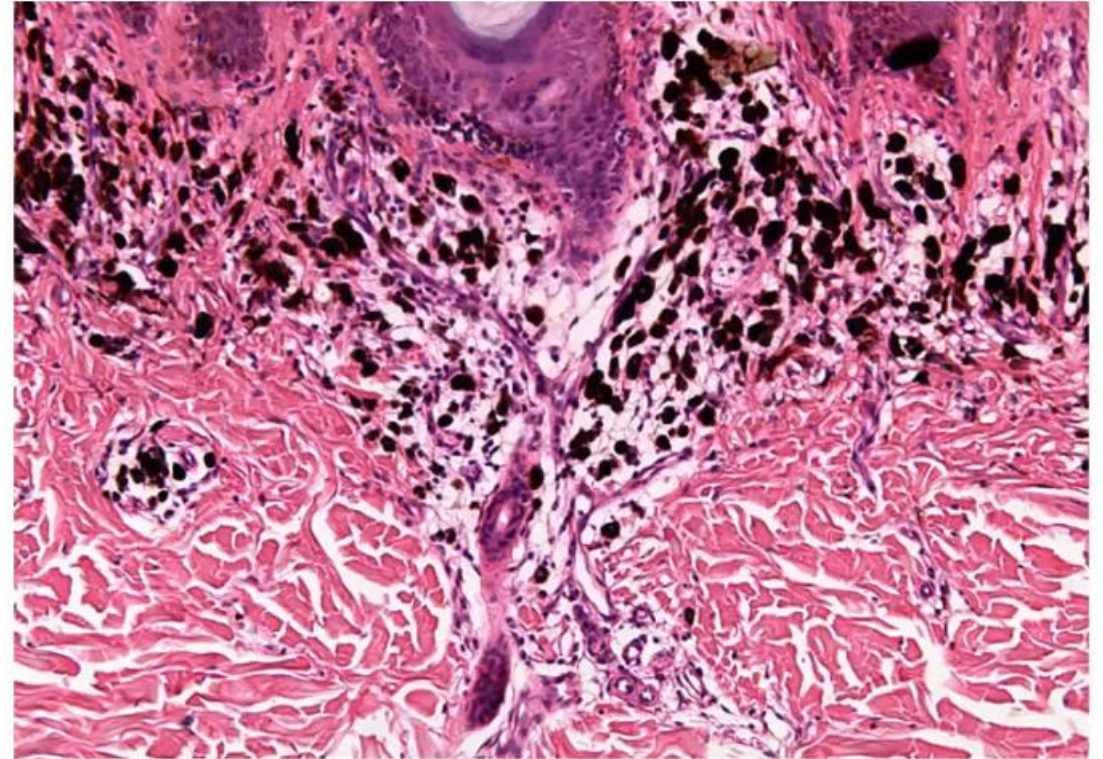
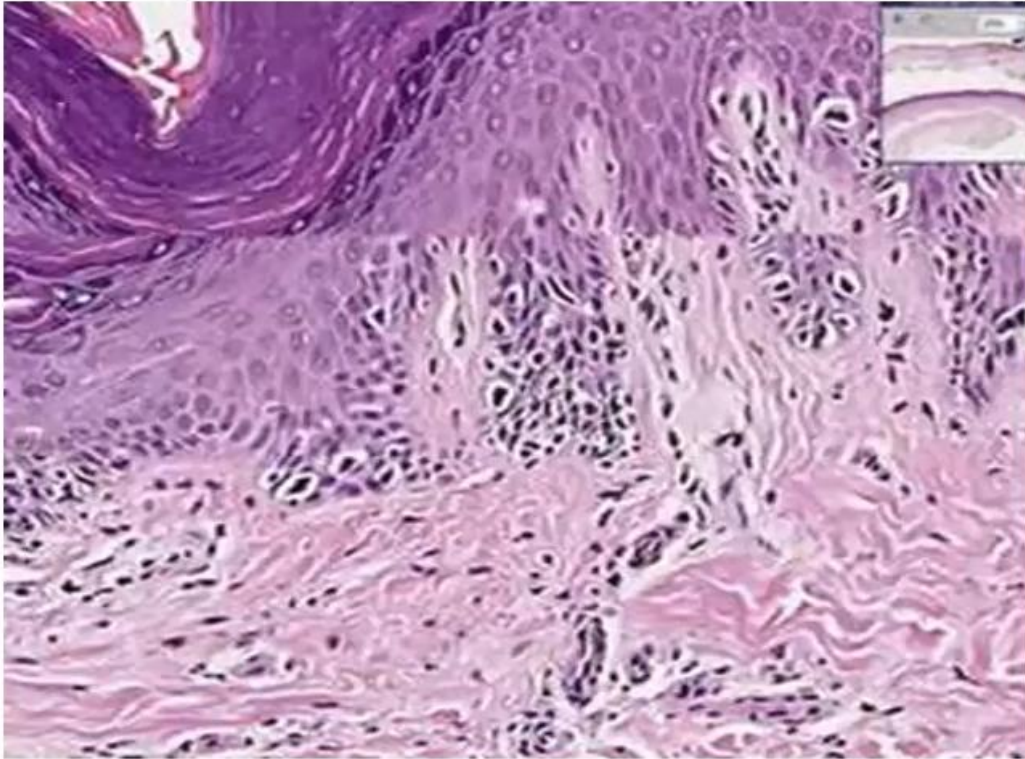


Clark Levels

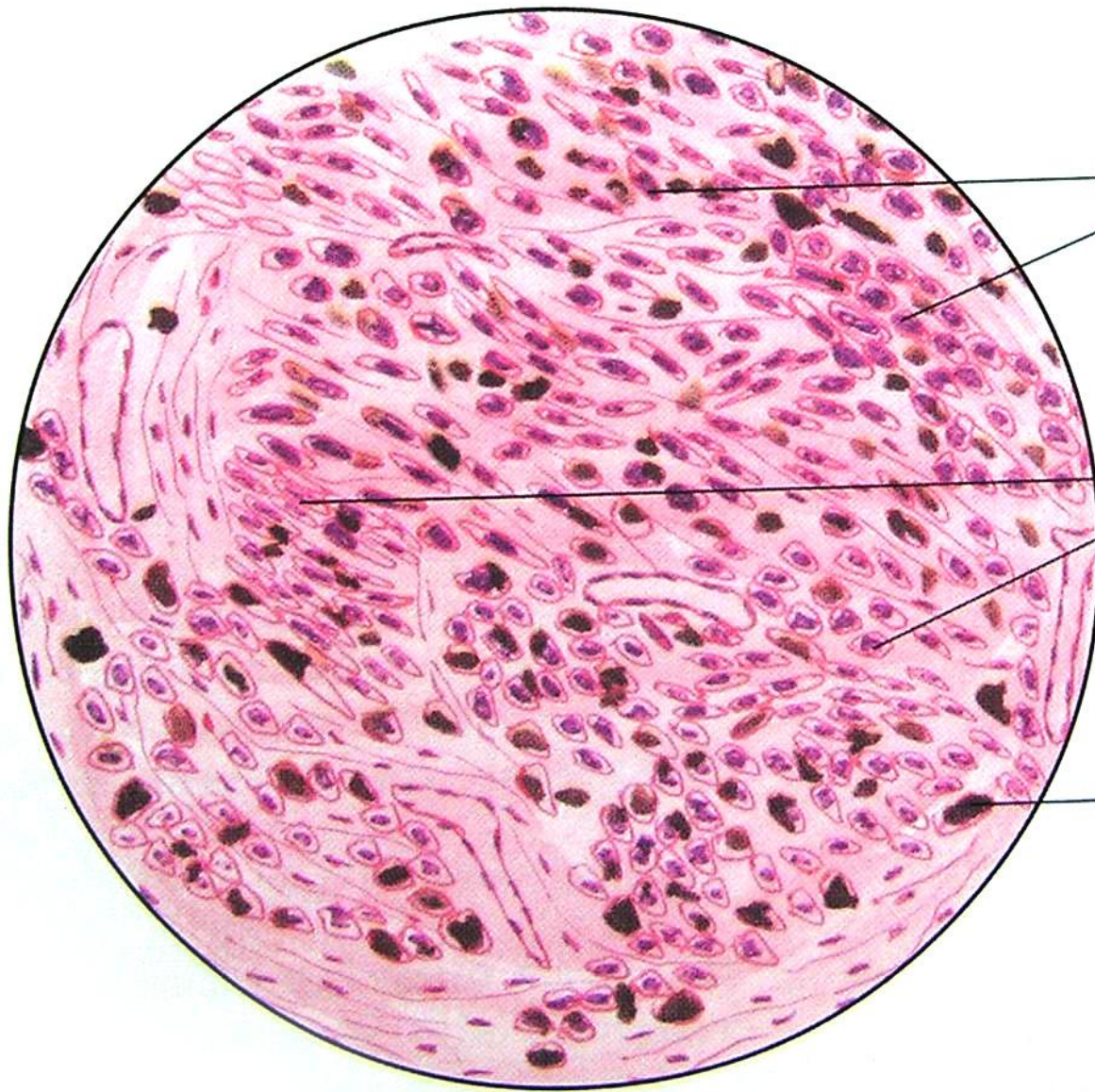


© 2009 Terese Winslow
U.S. Govt. has certain rights

Histopathology



- Dysplastic features in Melanocytes/Nevus cells



Malignant melanocytes
invading connective tissue

Loosely arranged sheets
and cords of epithelioid and
spindle melanocytes

Melanin pigments

Malignant melanoma

Immunohistochemical Markers:

- **S-100**: 90% positivity.
- **HMB-45**: 95.5% positivity.
- **Melan-A**: 100% positivity

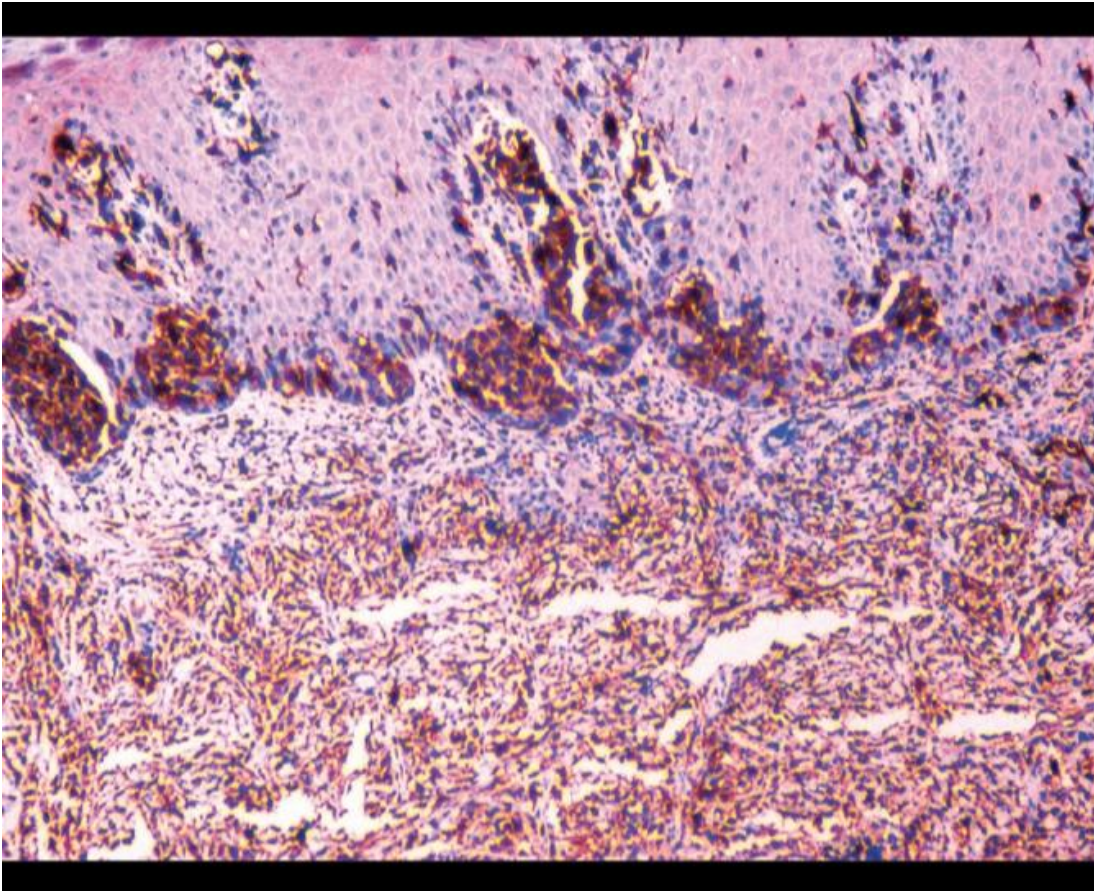


Figure 4

Immunohistochemistry showing strong reactivity of tumor cells to S-100 [x4]

Source

Oral malignant melanoma –
Clinical presentation with
immunohistochemical analysis

Journal of Cancer Research and
Therapeutics19(Suppl 2):S935-
S938, 2023.

References

1. Sheffer, J.E.S. (2015). *Oral Pathology: A Comprehensive Atlas and Text*.
2. Regezi, J.A., Sciubba, J.J., & Jordan, R.C.K. (2017). *Oral Pathology: Clinical Pathologic Correlations*.
3. Laimer, J., et al. (2018). Amalgam tattoo versus melanocytic neoplasm - Differential diagnosis of dark pigmented